

LEGAL AGCM, AB, Canada

WMO: 715440

Lat: 54.0031N

Lon: 113.4744W

Elev: 2231

StdP: 13.55

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.4	-17.7	-29.5	1.1	-22.9	-23.5	1.6	-17.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	81.4	62.8	78.6	61.5	75.8	60.5	66.6	76.7	64.5	74.5	62.5	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	93.0	72.1	60.6	85.7	69.7	58.5	79.6	67.0	32.5	76.5	30.8	74.4	29.3	72.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-28.8	87.4	2.8	2.1	-30.8	88.8	-32.5	90.1	-34.0	91.2	-36.1	92.7
			WB	-29.0	71.1	2.8	2.5	-31.0	72.9	-32.6	74.4	-34.2	75.8	-36.2	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	13.7	12.6	23.3	38.2	51.8	58.8	62.3	60.8	51.9	39.4	23.8	11.4		
	DBStd	21.76	15.74	15.00	13.91	9.27	8.42	5.34	4.59	5.77	8.11	8.79	12.44	14.25		
	HDD50	5852	1125	1048	829	364	80	5	0	3	72	343	785	1198		
	HDD65	10113	1590	1468	1294	803	412	197	109	153	395	794	1235	1663		
	CDD50	1279	0	0	0	12	137	268	382	337	129	14	0	0		
	CDD65	65	0	0	0	0	4	10	26	23	2	0	0	0		
	CDH74	988	0	0	0	4	113	140	326	302	101	2	0	0		
	CDH80	151	0	0	0	0	16	18	44	53	20	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	17.0	0.9	0.4	0.7	1.0	1.9	3.2	3.2	2.2	1.4	0.7	0.8	0.7		
	PrecMax	21.2	3.7	0.9	1.5	1.9	3.1	5.3	6.3	4.6	3.1	1.3	2.8	1.3		
	PrecMin	12.6	0.0	0.0	0.1	0.0	0.3	1.3	1.5	0.9	0.2	0.1	0.1	0.0		
	PrecStd	2.8	0.9	0.3	0.4	0.5	0.7	1.1	1.2	0.9	0.7	0.4	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	44.6	52.4	72.7	82.1	82.7	84.5	85.3	83.1	70.9	54.3	42.0		
		MCWB	36.7	37.2	43.0	51.5	55.7	62.4	67.6	63.1	60.4	53.8	44.8	35.6		
	2%	DB	40.9	39.4	46.0	64.3	77.3	78.1	80.5	80.6	76.4	62.0	46.4	38.6		
		MCWB	35.5	34.2	38.5	46.7	54.0	60.2	65.1	64.0	58.3	48.6	40.1	33.5		
	5%	DB	38.7	36.5	42.5	58.2	72.5	74.4	77.7	77.2	72.0	57.1	42.4	35.3		
		MCWB	34.3	32.5	36.7	43.9	52.2	58.6	63.7	62.8	56.0	45.8	36.9	31.1		
	10%	DB	35.7	33.2	39.1	53.6	68.3	71.3	74.6	74.1	66.7	52.7	38.4	31.2		
		MCWB	32.2	30.0	34.2	41.6	50.3	57.2	62.1	61.2	54.1	43.7	34.9	27.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	37.8	43.4	51.7	57.7	65.4	70.1	69.0	62.3	54.5	45.4	36.5		
		MCDB	42.2	43.4	51.2	71.4	73.6	77.5	80.2	79.1	77.2	68.0	53.4	41.1		
	2%	WB	36.0	35.0	39.4	47.4	55.7	62.4	67.5	66.3	59.4	50.2	40.4	33.8		
		MCDB	40.2	38.9	45.0	62.4	72.0	73.6	77.5	76.9	73.4	59.6	45.5	38.0		
	5%	WB	34.6	32.5	36.8	44.7	53.8	60.4	65.3	64.1	57.2	47.0	37.5	31.1		
		MCDB	38.4	35.9	42.1	56.8	69.6	71.3	74.8	74.1	69.8	55.3	41.2	35.2		
	10%	WB	32.1	30.2	34.6	42.4	51.9	58.7	63.3	62.1	55.1	44.6	35.0	27.7		
		MCDB	35.9	33.9	38.7	52.0	65.0	68.5	72.3	71.7	65.3	51.3	38.3	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	16.0	15.9	18.6	22.4	19.6	20.7	21.6	21.2	17.3	13.7	14.3		
		MCDBR	13.2	16.5	17.8	27.5	28.8	24.8	25.7	26.4	30.5	24.2	17.4	17.5		
		MCWBR	10.4	13.1	11.9	14.1	12.0	11.1	13.2	12.4	14.2	13.7	11.9	14.3		
	5% WB	MCDBR	13.3	15.8	17.3	25.8	26.0	21.7	23.7	24.2	28.6	23.2	15.8	17.3		
		MCWBR	10.9	12.9	11.9	13.8	11.2	10.7	13.2	12.5	13.8	13.5	11.6	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.270	0.308	0.324	0.339	0.351	0.352	0.359	0.305	0.280	0.251	0.236		
	taud		2.294	2.325	2.301	2.392	2.399	2.377	2.420	2.350	2.507	2.549	2.453	2.310		
	Ebn at Noon		230	266	277	288	288	285	282	272	278	264	236	216		
	Edh at Noon		18	24	31	33	35	36	34	34	25	20	15	14		
(44) All-Sky Solar Radiation	RadAvg		288	615	1099	1479	1808	1896	1927	1583	1092	620	292	210		
	RadStd		33	46	83	130	142	125	99	111	112	78	32	19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page